III. Existing Systems/State of the Art

1. Overview of the Field

Shrimp aquaculture has repeatedly been affected by viral, bacterial, parasitic, and environmentally mediated disease events. Historical reviews describe how pathogens such as WSSV, IHHNV, YHV, and related syndromes caused severe production losses and forced the industry to adopt stronger biosecurity, surveillance, and health-management practices [1–9]. These studies establish the practical motivation for rapid screening, but they also show why image analysis cannot replace laboratory confirmation: multiple diseases can share visible symptoms, signs may appear only at particular infection stages, and environmental stress can produce visually similar abnormalities.

Computer vision provides a complementary, non-invasive route for preliminary screening. In the broader smart-aquaculture literature, deep learning has been applied to species recognition, biomass estimation, behavior monitoring, water-quality support, and disease-oriented image analysis [10]. For shrimp disease research, the most relevant tasks are image classification and instance segmentation. Classification assigns an image-level label and is appropriate for fast mobile screening. Instance segmentation predicts a separate mask for each visible disease region and therefore supports spatial evidence, stricter localization evaluation, and investigation of small or irregular lesions.

The literature is fragmented across several technical directions. Shrimp-specific studies often optimize clean-test classification accuracy on a single dataset, whereas general computer-vision research separately advances efficient backbones, imbalanced learning, attention, corruption robustness, explainable AI, instance segmentation, and mobile deployment. A defensible state-of-the-art review must therefore connect these directions without treating metrics from different datasets as directly rankable. Reported results are meaningful only together with the dataset, class definitions, preprocessing, split policy, checkpoint selection, and evaluation metric used by the original study.

This chapter follows a chronological and analytical structure. It first traces the transition from biological disease management to image-based screening, then examines the theories and systems that support lightweight classification and mask-level localization. The comparison tables preserve results and claims from the reviewed publications while explicitly identifying limitations relative to the thesis. Project-specific proposed modules and experimental results are not presented as existing methods; they are introduced only in the later Methodology and Results chapters.

2. Historical Context

2.1 Biological and Conventional Diagnostic Foundations

Before deep learning became relevant, shrimp disease research concentrated on pathogen emergence, transmission, clinical signs, histopathology, molecular diagnosis, and farm-level control. Studies from the late 1990s onward documented the destructive effect of viral diseases on penaeid shrimp and emphasized exclusion, pathogen-free stocks, movement control, and laboratory testing [1], [2]. Later reviews connected recurrent disease emergence with the rapid globalization and intensification of shrimp production [3], [4].

By 2012, the literature had shifted from describing individual outbreaks toward a system-level view of food security and pathogen dissemination. Stentiford et al. argued that disease could constrain future crustacean supply, while Flegel and Seibert and Pinto reviewed the emergence, distribution, and biology of major shrimp pathogens [5–7]. Thitamadee et al. subsequently summarized important disease threats facing cultivated penaeid shrimp in Asia, including syndromes whose visible signs can overlap with environmental stress or secondary infection [8]. This history explains why an image model should be framed as preliminary screening rather than autonomous diagnosis.

2.2 Early Image Analysis and Transfer Learning

The first practical transition toward data-driven image recognition occurred when convolutional neural networks became transferable to small domain datasets. Duong-Trung et al. demonstrated that pretrained CNNs could classify multiple shrimp disease categories, establishing an early field-image baseline for this topic [11]. The study was important because it reduced dependence on handcrafted visual descriptors, but it remained centered on image-level prediction and did not provide disease-region masks, mobile deployment evidence, or specimen-grouped leakage analysis.

A second stage emphasized compact custom architectures. Wang et al. modified LeNet for rapid classification of hepatopancreatic necrosis, red-body, and whitish-muscle disease in Penaeus vannamei [12]. This work illustrated that shrimp-specific simplification could reduce computational cost, although its disease space and acquisition protocol differ from four-class Healthy/BG/WSSV/co-infection screening. In parallel, Querol et al. investigated MobileNetV3-Small and EfficientNetV2-B0 for WSSV monitoring, combined on-device considerations with saliency heatmaps, and explicitly discussed class imbalance [13]. The progression was therefore not merely from one architecture to another; it was a movement from offline classification toward resource-aware and interpretable screening.

2.3 Public Datasets and Data-Centric Development

The field expanded more rapidly after the release of reusable image resources. ShrimpDiseaseBD and its associated ShrimpDiseaseImageBD record introduced a four-class collection containing Healthy, BG, WSSV, and combined BG/WSSV images collected with farm collaboration and expert supervision [14], [15]. TigerShrimpBD provided an additional public source that was later used together with ShrimpDiseaseImageBD by a lightweight classification study [16], [17]. These releases made independent benchmarking more feasible, but image-level labels did not remove the need to audit repeated specimens, near duplicates, acquisition bias, or annotation uncertainty.

Adjacent aquatic-disease datasets are useful only when their domain difference is explicit. SalmonScan, for example, supports fish-disease image research but differs in species, symptom morphology, acquisition conditions, and class semantics [18]. Such datasets can demonstrate broader trends in aquatic computer vision, but they should not be treated as direct substitutes for shrimp-specific validation.

2.4 Lightweight, Explainable, and Deployable Systems

From 2024 onward, shrimp disease research increasingly combined classification with application-oriented requirements. SHRIMPAI implemented a mobile WSSV screening workflow

[19], while Xu et al. and Fei et al. explored YOLO-based shrimp disease localization or detection [20], [21]. Ramachandran et al. combined an enhanced gated recurrent unit with wild-geese-migration optimization for WSSV-oriented shrimp classification [22]. These studies broadened the field from generic transfer learning toward disease-specific deployment and localization, although object detection remained different from instance segmentation because bounding boxes do not describe the exact disease-region boundary.

The 2025–2026 literature further diversified. Büyükarıkan evaluated multi-model ensembles; Aung et al. compared MobileNet and EfficientNet in a pathogen-oriented imaging setting; Raj et al. introduced a recurrent capsule architecture; and other studies combined classification with recommendations, IoT monitoring, or severity assessment [23–28]. ShrimpXNet is the closest classification reference for the present thesis because it combines background removal, transfer learning, MixUp, CutMix, adversarial training, and CAM-based explanation [29]. FeatherNetX represents a different direction: a compact architecture evaluated through five-fold cross-validation and two public shrimp image sources [17]. Recent MobileNetV3–KNN and explainable ensemble studies continue this trend toward compact or interpretable systems, while additional conference work has continued the direct shrimp-disease classification line [30–32].

3. Key Studies and Theories

3.1 Shrimp Disease Classification

Shrimp disease classification studies differ substantially in label space. Some use six disease categories, some focus only on WSSV, and others classify pathological tissue or a small set of visibly distinct syndromes. Duong-Trung et al. reported an early multi-disease transfer-learning result, whereas Wang et al. designed a compact LeNet derivative for three Penaeus vannamei diseases [11], [12]. WSSV-focused systems subsequently evaluated mobile backbones or application workflows [13], [19], [22]. These studies demonstrate feasibility, but their numerical results cannot be placed in a single ranking because the datasets and target definitions are not equivalent.

The more recent four-class literature is more directly relevant. ShrimpXNet evaluates several pretrained backbones on the public four-class dataset and incorporates regularization, adversarial training, and explanation methods [29]. FeatherNetX combines a compact backbone with public shrimp datasets and reports model size and computational complexity in addition to classification performance [17]. Büyükarıkan, Raj et al., and the explainable ensemble study demonstrate alternative ensemble, capsule, and multi-backbone directions [23], [25], [31]. Together, these papers show that the field has moved beyond asking whether CNNs can recognize shrimp disease; current questions concern efficiency, robustness, class imbalance, interpretability, external evaluation, and reproducibility.

ShrimpXNet must nevertheless be described carefully. It is a literature baseline and methodological reference for classification, not the architecture directly modified by the proposed classifier in this thesis. Its lack of a complete official reproduction package also means that any local reimplementation must document data protocol, preprocessing, augmentation, architecture, optimization, checkpoint selection, evaluation definitions, framework versions, and nondeterministic behavior rather than attributing differences only to a generic “different environment.”

3.2 Public Shrimp and Aquatic-Disease Datasets

Dataset papers make the experimental object explicit and therefore deserve separate analysis from model papers. ShrimpDiseaseBD documents the acquisition and class organization of the Bangladesh shrimp disease images, while the Mendeley record preserves the downloadable dataset version and DOI [14], [15]. TigerShrimpBD is relevant because it was used in later peer-reviewed lightweight classification work [16], [17]. The main limitation of these sources for instance segmentation is that they provide image-level disease categories rather than independently reviewed disease-region masks.

A dataset description is not sufficient evidence of split validity. Multiple views of the same physical specimen can produce highly similar images with consistent disease labels. If such views cross training and test partitions, the model may exploit specimen-specific appearance rather than disease-general evidence. Dataset comparison must therefore include label type, annotation provenance, source identity, repeated-subject risk, and whether the original publication reports a grouped or source-aware split.

3.3 Instance-Segmentation Foundations

Dense prediction developed through several architectural stages. Fully convolutional networks and U-Net established pixel-wise semantic prediction and encoder–decoder recovery of spatial detail [33], [34]. Mask R-CNN then extended two-stage detection with a parallel mask branch, providing a foundational framework for instance-level masks [35]. YOLACT separated prototype-mask generation from instance-specific coefficients, showing that real-time instance segmentation could be obtained with a one-stage formulation [36].

Later systems targeted boundary quality, ROI-free prediction, dynamic masks, and real-time query processing. PointRend treated segmentation as iterative rendering of uncertain points; CondInst generated instance-specific dynamic mask heads without ROI operations; SOLOv2 represented instances by location and dynamic kernels; and Mask2Former unified several segmentation tasks through masked attention [37–40]. YOLACTEdge, FastInst, and MobileInst moved the efficiency discussion closer to edge or mobile constraints [41–43]. These methods are not shrimp-specific, but they establish the design space for disease-region masks and clarify why classification accuracy alone cannot validate localization quality.

3.4 Lightweight Classification Architectures

The classification baseline families benchmarked by this project originate from a long sequence of architectural improvements. AlexNet, VGG, GoogLeNet, ResNet, DenseNet, and Xception established progressively deeper, multi-branch, residual, densely connected, and depthwise-separable representations [44–49]. SqueezeNet and MobileNet then made parameter count and deployment cost explicit design objectives [50], [51].

Mobile-oriented research subsequently introduced channel shuffling, inverted residuals, platform-aware search, hardware-aware activations, compound scaling, cheap feature generation, reparameterization, and hybrid CNN–transformer designs [52–65]. These papers provide the original architectural context for the TIMM screening table. Their relevance is not that every model is suitable for the final application, but that they allow the thesis to compare accuracy and Macro-F1 against parameter count, model size, latency, and conversion feasibility.

3.5 Imbalance-Aware Learning

Disease datasets are often imbalanced because Healthy images and common disease classes may be easier to collect than mixed or less frequent conditions. Focal Loss reduces the influence of easy examples; Class-Balanced Loss reweights classes using the effective number of samples; LDAM introduces class-dependent margins; and Asymmetric Loss applies different focusing behavior to positive and negative examples [66–69]. These are existing published methods and can be compared theoretically. The project-specific ASL–LDAM formulation, however, is not an existing related-work method and is therefore introduced only in the Methodology chapter.

The literature also warns against assuming that one imbalance method is universally superior. Loss behavior depends on whether the task is multiclass or multilabel, the severity of imbalance, label noise, calibration, sampling strategy, and model capacity. A fair thesis comparison therefore keeps the split, backbone, training schedule, augmentation, checkpoint selection, and metric definitions controlled when evaluating an imbalance-aware objective.

3.6 Attention and Feature Recalibration

Attention modules seek to redistribute representational capacity toward informative channels or spatial regions. Squeeze-and-Excitation models channel dependencies; CBAM applies sequential channel and spatial attention; ECA reduces channel-attention overhead; Coordinate Attention preserves positional information; SimAM derives parameter-free neuron importance; and Triplet Attention models cross-dimensional interactions [70–75]. These methods provide published foundations for controlled attention experiments in segmentation.

The review does not classify the project-specific attention configuration or DCFR module as an existing system. DCFR is a custom, independent feature-recalibration component, and its design and evidence belong to the proposed methodology. Likewise, an attention configuration selected through the project's instance segmentation experiments is a proposed configuration rather than a peer-reviewed baseline. Related work should explain the adopted foundations without assigning publication status to the project's own design.

3.7 Fourier-Domain Processing for Segmentation

Frequency-domain research provides a theoretical basis for examining image degradation and texture-sensitive segmentation. Yin et al. showed that robustness trade-offs can be interpreted through the frequency concentration of corruptions and augmentation effects [76]. Fourier Domain Adaptation transferred low-frequency amplitude information between source and target images for semantic segmentation, while a later Fourier framework investigated domain generalization through frequency manipulation [77], [78].

These studies motivate three segmentation-specific questions: whether Fourier transformations improve robustness under task-appropriate degradation; whether they preserve or enhance evidence for small disease regions; and whether they produce measurable benefits for instance segmentation. Existing literature does not answer these questions for BG/WSSV masks. The project's Fourier configurations and experimental answers are therefore evaluated later as proposed research, not listed here as published existing methods.

3.8 Robustness and Explainable AI

Robustness research distinguishes clean-test performance from stability under distribution shift. FGSM established a simple adversarial perturbation mechanism; mixup and CutMix regularized training through sample interpolation or region replacement; ImageNet-C formalized common-corruption benchmarking; and AugMix improved corruption robustness through diverse augmentation chains [79–83]. These methods support a controlled robustness protocol but do not justify claiming resilience to every farm condition.

Explainability research progressed from class activation mapping to gradient- and score-based variants. CAM links global-average-pooled features to class scores; Grad-CAM and Grad-CAM++ produce class-discriminative heatmaps; Score-CAM weights activation maps using forward scores; and Eigen-CAM produces a class-independent principal-component visualization [84–88]. Heatmaps can identify influential regions and expose obvious background dependence, but they do not establish causality, clinical validity, or pixel-accurate disease boundaries. This distinction is important when comparing XAI with instance segmentation.

3.9 Leakage-Aware and Reproducible Evaluation

Data leakage occurs when training or model-selection procedures gain information that would not be available in deployment. Kaufman et al. formalized leakage as an evaluation-design problem rather than merely a coding error [89]. In medical imaging, Zech et al. showed that site-specific confounding can weaken cross-domain generalization; Roberts et al. documented recurring methodological pitfalls; and Tampu et al. quantified inflated test accuracy caused by leakage [90–92].

The analogous shrimp-dataset risk is repeated-specimen leakage. If several photographs of the same shrimp appear in different partitions, an image-level random split can overestimate generalization. A specimen-grouped split reduces this specific risk by keeping all known views of one specimen together. It does not eliminate label error, acquisition-site bias, temporal confounding, or undocumented duplicates. Reproducible evaluation therefore requires split manifests, stable class ordering, identifiable checkpoints, recorded seeds, saved configurations, and traceable metric scripts.

3.10 Mobile and Edge Deployment

Mobile deployment requires more than a low parameter count. Model operators must be supported by the target runtime, preprocessing must match training, tensor order and class mapping must remain stable, and latency must be measured on representative hardware. MobileNet and later mobile architectures make hardware efficiency an explicit design objective [51], [53], [55], [65]. Shrimp-specific edge and mobile studies demonstrate the practical value of on-device or near-device screening [13], [19], [27].

The recent YOLO lineage also matters because unified implementations can support classification and instance segmentation within one tooling ecosystem. The original YOLO paper established single-stage real-time detection, YOLOv10 advanced end-to-end NMS-free detection, and the YOLO26 technical paper describes a unified family covering classification and instance segmentation [93–95]. Object-detection results are not treated as a research outcome of this thesis; the relevance is the shared deployment-oriented framework and the availability of lightweight classification and segmentation variants.

4. Technological Advancements

4.1 From Manual Inspection to Image-Based Screening

The earliest stage relied on visual observation supported by microscopy, histopathology, and molecular tests. Image-based screening does not remove the need for those procedures, but it can prioritize suspicious cases and shorten the time between observation and expert follow-up. The responsible technological shift is therefore from unaided observation to decision support, not from laboratory diagnosis to autonomous clinical judgment.

4.2 From Handcrafted Features to Transfer Learning

Transferred CNNs reduced the need to design disease-specific color, texture, and shape descriptors manually. Early shrimp studies demonstrated feasibility with pretrained networks, while later work used compact custom models, ensembles, capsule networks, and hybrid pipelines [11], [12], [23], [25]. The principal advancement was reusable representation learning, but model comparison remained sensitive to dataset construction and split policy.

4.3 From Heavy CNNs to Lightweight Architectures

The efficiency literature replaced raw depth with structured operations such as depthwise convolution, channel shuffle, inverted residuals, cheap feature generation, neural architecture search, and structural reparameterization [52–55], [57], [63]. This progression supports multi-objective selection using Macro-F1, parameters, size, latency, and export feasibility. A heavier model is not automatically preferable when its performance gain is small or absent.

4.4 From Image-Level Labels to Instance-Level Evidence

Classification answers which class best represents an image, whereas instance segmentation asks where each disease region is located. The technological path from FCN and U-Net to Mask R-CNN, YOLACT, CondInst, Mask2Former, and mobile-oriented systems gradually improved instance separation, boundary representation, speed, and deployment feasibility [33–36], [38], [40], [43]. For shrimp images, the principal barrier is not only architecture selection but the absence of expert-reviewed disease-region masks in the public classification dataset.

4.5 From Clean Accuracy to Robust and Transparent Evaluation

Recent systems increasingly supplement accuracy with class-balanced metrics, corruption tests, adversarial evaluation, and explanation maps. This shift is important because a classifier can achieve high clean accuracy while relying on acquisition artifacts or majority-class shortcuts. Robustness and XAI provide complementary evidence: robustness measures stability under controlled transformations, whereas heatmaps help inspect influential image regions. Neither replaces external validation or mask-level localization.

4.6 From Random Splits to Leakage-Aware Protocols

Random image splitting is convenient but scientifically weak when images are clustered by specimen, source, or acquisition session. The evaluation literature has progressively emphasized subject-level separation, site-aware validation, and transparent reporting [89–92]. In a shrimp dataset with repeated photographs, grouped-specimen splitting is the appropriate direct response, while source-wise splitting before multi-dataset merging protects partition integrity.

4.7 From Offline Models to Integrated Mobile Systems

The final technological shift joins research evidence with application delivery. SHRIMPAI, the edge-WSSV study, and the integrated IoT system show that disease models can be embedded in mobile or edge workflows [13], [19], [27]. A complete implementation must preserve preprocessing, class mapping, model metadata, result interpretation, and data-storage behavior. Deployment should therefore be evaluated as a reproducible pipeline rather than a screenshot of a successful prediction.

5. Comparison of Existing Systems

The following tables organize the literature by chronology, task, dataset, evaluation coverage, and limitation. Numerical results are preserved only when the original source reports them clearly. They are not treated as directly comparable rankings because the studies differ in species, disease definitions, image modality, dataset size, split policy, augmentation, architecture, and evaluation metric.

Table 3.1. Chronological evolution of related work

Year	Study / method	Task and main contribution	Relevance to this thesis
1997	Flegel [1]	Shrimp disease review: Summarized major viral diseases affecting black tiger shrimp.	Establishes the biological and economic context.
1998	Lightner [2]	Disease control: Reviewed practical viral-disease control strategies.	Defines conventional management boundaries.
2009	Walker and Mohan [3]	Disease emergence: Connected viral emergence, impact, and health management.	Motivates early screening and surveillance.
2011	Lightner [4]	Shrimp virology review: Reviewed farmed-shrimp virus diseases in the Americas.	Shows geographic and pathogen diversity.
2012	Stentiford et al. [5]	Food-supply risk: Linked crustacean disease to future seafood supply.	Supports project significance.
2012	Flegel; Seibert and Pinto [6], [7]	Pathogen history: Reviewed major shrimp pathogens and viral dispersion.	Explains why visible symptoms require cautious interpretation.

Year	Study / method	Task and main contribution	Relevance to this thesis
2016	Thitamadee et al. [8]	Disease-threat review: Synthesized current threats for cultivated penaeid shrimp.	Defines realistic disease complexity.
2018	Shinn et al. [9]	Economic impact: Assessed disease-related production costs in Asian shrimp farming.	Strengthens practical motivation.
2015–2017	FCN, U-Net, Mask R-CNN [33–35]	Segmentation foundations: Advanced pixel-wise and instance-level mask prediction.	Provides mask-localization foundations.
2016–2019	YOLO, YOLACT [36], [93]	Real-time vision: Moved detection and instance segmentation toward real-time inference.	Supports lightweight localization context.
2017–2024	Mobile architecture families [51–53], [55], [65]	Mobile classification: Designed efficient operators and hardware-aware models.	Supports edge-oriented model selection.
2017–2021	Focal, class-balanced, LDAM, ASL [66–69]	Imbalanced learning: Introduced focusing, reweighting, margin, and asymmetric objectives.	Provides imbalance-learning foundations.
2016–2020	CAM families [84–88]	Explainable AI: Produced visual evidence from classifier activations.	Supports qualitative interpretation.
2018–2020	mixup, CutMix, ImageNet-C, AugMix [80–83]	Robustness: Expanded regularization and common-corruption evaluation.	Supports robustness protocol design.
2019–2021	Fourier robustness and adaptation [76–78]	Frequency-domain learning: Connected frequency content with robustness and domain shift.	Motivates Fourier segmentation experiments.
2020	Duong-Trung et al. [11]	Shrimp classification: Applied transferred CNNs to multi-disease shrimp images.	Early direct classification baseline.
2020–2024	CondInst, Mask2Former, MobileInst [38], [40], [43]	Efficient instance segmentation: Improved dynamic, query-based, and mobile-oriented masks.	Defines modern segmentation design space.
2023	Wang et al. [12]	Compact shrimp classification: Modified LeNet for rapid Penaeus vannamei disease recognition.	Shows shrimp-specific lightweight design.
2023	Querol et al. [13]	Edge WSSV recognition: Compared mobile backbones, imbalance handling, and saliency.	Closest early edge/XAI direction.
2024	SHRIMPAI [19]	Mobile WSSV application: Integrated CNN screening into a mobile application.	Provides mobile workflow precedent.

Year	Study / method	Task and main contribution	Relevance to this thesis
2024	Xu et al. [20]	Shrimp disease detection: Combined improved YOLOv8 with multiple visual features.	Supports localization context, not mask evidence.
2024	Ramachandran et al. [22]	WSSV classification: Combined an enhanced GRU with wild-geese-migrati on optimization.	Adds disease-specific sequence-modeling evidence.
2025	ShrimpDiseaseBD/ ImageBD [14], [15]	Public dataset: Released a documented four-class shrimp-disease image source.	Primary classification dataset reference.
2025	TigerShrimpBD [16]	Public dataset: Released an additional tiger-shrimp image source.	Supports later multi-source evaluation.
2025	Büyükarıkan [23]	CNN ensemble: Evaluated multi-model ensemble strategies.	Shows ensemble performance and complexity trade-offs.
2025	Aung et al. [24]	Pathogen-image classification: Compared MobileNet and EfficientNet in a different imaging setting.	Adjacent evidence; not a direct smartphone baseline.
2025	Raj et al. [25]	Capsule/recurrent classification: Combined capsule, recurrent, attention, and optimization components.	Alternative architecture with high complexity.
2025	Fei et al. [21]	Lightweight detection: Reduced YOLOv8n complexity for shrimp disease detection.	Supports efficient localization context.
2026	ShrimpXNet [29]	Robust/XAI classification: Combined preprocessing, augmentation, adversarial training, and CAM.	Closest literature classification baseline.
2026	FeatherNetX [17]	Lightweight classification: Reported a compact model with cross-validation and two data sources.	Strong efficiency and reproducibility reference.
2026	MobileNetV3–KNN [30]	Early detection: Integrated a mobile CNN with KNN classification.	Alternative compact deployment approach.
2026	Explainable ensemble CNN [31]	Explainable classification: Combined multiple backbones, imbalance handling, and Grad-CAM.	Recent direct XAI-oriented reference.
2026	IoT–ResNet monitoring [27]	Integrated monitoring: Combined image classification with pond sensing and edge inference.	Shows system-level deployment integration.
2026	YOLO26 [95]	Unified vision family: Provides classification and instance segmentation variants in one pipeline.	Relevant engineering backbone for classification and instance segmentation variants.

Table 3.1 shows a continuous change in research emphasis. The earliest literature defined the biological problem and the limits of conventional disease management. General computer vision then provided classification, segmentation, robustness, explainability, and mobile-efficiency foundations. Shrimp-specific studies first adopted transfer learning, subsequently introduced compact architectures and mobile workflows, and only recently began combining public datasets, robustness, XAI, external sources, or integrated sensing. This chronology also exposes an imbalance

in the literature: classification systems are increasingly mature, whereas peer-reviewed shrimp disease instance segmentation and leakage-aware specimen splitting remain scarce.

Table 3.2. Representative shrimp disease classification systems

Study	Dataset and method	Reported evidence and strength	Limitation for this project
Duong-Trung et al. [11]	Study-specific six-disease image set; Transferred CNN classification	90.02% accuracy reported; Early direct field-image study	No masks, mobile validation, or grouped split reported
Wang et al. [12]	Three Penaeus vannamei disease classes; Improved LeNet	Rapid compact recognition reported; Shrimp-specific lightweight design	Different classes and protocol from this thesis
Querol et al. [13]	WSSV monitoring dataset; MobileNetV3-Small / EfficientNetV2-B0	F1 = 0.72 / 0.99; Edge focus, imbalance discussion, saliency	Binary task and no mask evaluation
SHRIMPAI [19]	WSSV mobile screening; Sequential CNN and mobile app	Successful tests on two shrimp species reported; Complete application-oriented workflow	Limited sample scale and binary disease focus
Ramachandran et al. [22]	WSSV-oriented shrimp images; Enhanced GRU with wild-geese optimization	Improved WSSV-classification performance reported; Disease-specific optimized recurrent model	Single-disease setting
Büyükarıkan [23]	ShrimpDiseaseBD; Multi-model CNN ensemble	Study hypothesis and results exceed 0.90 accuracy; Strong ensemble comparison	Higher complexity; no mask localization
Aung et al. [24]	Pathogen/histopathology images; MobileNet and EfficientNet	Accuracy above 95% reported; Lightweight pathology recognition	Different modality from farm smartphone images
Raj et al. [25]	Shrimp disease images; Recurrent capsule and hybrid optimization	Precision about 94.9% reported; Rich spatial/ channel modeling	Complex pipeline and limited deployment evidence
ShrimpXNet [29]	ShrimpDiseaseImageBD; four classes; Transfer learning, preprocessing, MixUp/CutMix, FGSM, CAM	ConvNeXt-Tiny 96.88% test accuracy reported; Closest robustness/XAI classification reference	Reproduction code is not fully specified

Study	Dataset and method	Reported evidence and strength	Limitation for this project
FeatherNetX [17]	ShrimpDiseaseImageBD + TigerShrimpBD; Compact custom classifier; five-fold CV	93% average accuracy; 0.739M parameters; Strong efficiency reporting and multi-source use	Desktop deployment; no instance masks
Explainable ensemble [31]	Enhanced aquaculture image data; DenseNet121 + ConvNeXt-Tiny + SE-ResNeXt50	96.43% precision reported; Imbalance handling and Grad-CAM	Ensemble complexity and short conference report
IoT–ResNet system [27]	Four-class image set + pond sensors; Transfer-learned ResNet-50 and IoT monitoring	Accuracy 0.8559; F1 0.8552; edge latency reported; End-to-end monitoring context	No instance segmentation or leakage audit

Table 3.2 demonstrates why reported percentages must not be compared as if they came from a common benchmark. The studies vary from binary WSSV recognition to four- or six-class disease classification, and some use tissue or histopathology images rather than whole-shrimp farm photographs. ShrimpXNet is the closest literature reference because its public dataset, preprocessing, augmentation, adversarial training, and XAI overlap with the classification concerns of this thesis. FeatherNetX contributes stronger efficiency and multi-source evidence. Neither paper provides the mask-level BG/WSSV evaluation or specimen-grouped protocol required by the instance segmentation track.

Table 3.3. Localization, instance segmentation, and evaluation methods

Study / method	Task and main idea	Strength	Limitation
FCN / U-Net [33], [34]	Semantic segmentation; Encoder–decoder pixel prediction	Strong dense-prediction foundation	Does not inherently separate multiple instances
Mask R-CNN [35]	Instance segmentation; Two-stage detection with parallel mask branch	Accurate, established baseline	Higher computational and proposal-stage overhead
YOLACT [36]	Real-time instance segmentation; Prototype masks plus instance coefficients	Good speed–accuracy design	Prototype quality can limit fine boundaries
PointRend [37]	Boundary refinement; Iteratively predicts uncertain points	Improves high-resolution boundaries	Adds refinement complexity

Study / method	Task and main idea	Strength	Limitation
CondInst [38]	ROI-free instance segmentation; Dynamic instance-specific mask heads	High-resolution masks without ROI resizing	Detection-conditioned training remains complex
SOLOv2 [39]	One-stage instance segmentation; Location-based categories and dynamic kernels	Fast end-to-end formulation	Dense scene assumptions differ from disease regions
YOLACTEdge [41]	Edge instance segmentation; Edge-oriented YOLACT optimization	Direct edge deployment evidence	Video/edge design is not shrimp-specific
Mask2Former [40]	Universal segmentation; Masked-attention queries	Strong unified segmentation framework	Transformer complexity can conflict with mobile limits
FastInst / MobileInst [42], [43]	Real-time/mobile masks; Efficient query and mobile video designs	Demonstrate mobile-oriented dense prediction	Not validated for shrimp disease masks
Improved YOLO shrimp studies [20], [21]	Disease detection; Bounding-box localization with lightweight YOLO variants	Shrimp-specific localization evidence	Boxes do not provide disease-region boundaries
Fourier segmentation foundations [77], [78]	Domain adaptation/generalization; Frequency-amplitude manipulation	Motivates robustness and appearance studies	No direct BG/WSSV instance evidence
Leakage literature [89–92]	Evaluation validity; Detects leakage, confounding, subject overlap, and methodological pitfalls	Protects generalization estimates	Requires specimen/ source metadata that datasets may not provide

The localization literature develops from generic pixel prediction to explicit instance separation and then toward real-time or mobile operation. Shrimp-specific work is currently concentrated on bounding-box detection, while mask-level disease evidence depends mainly on general instance segmentation foundations. This gap justifies constructing BG/WSSV region masks, but it also imposes a quality limitation: a single-annotator mask set without independent expert review cannot be treated as a clinical ground truth.

Table 3.4. Public datasets relevant to shrimp and aquatic-disease vision

Dataset / source	Year and domain	Scale and labels	Relevance and limitation
ShrimpDiseaseImageBD [14], [15]	2025; Shrimp disease	1,149 original images; four classes; Image-level disease labels	Primary four-class classification source; no expert-reviewed instance masks
TigerShrimpBD [16]	2025; Tiger shrimp images	Public Mendeley dataset; Image-level categories	Used by FeatherNetX; supports multi-source classification analysis
SalmonScan [18]	2024; Salmon disease	Public Data in Brief resource; Image-level disease labels	Adjacent aquatic-disease evidence; species/domain differ
Study-specific WSSV sets [13], [19]	2023–2024; WSSV monitoring	Collected for edge/mobile studies; Binary image labels	Useful deployment precedent; limited reuse and disease diversity
Pathogen-image dataset [24]	2025; Shrimp pathogens	Curated pathology-oriented images; Pathogen/tissue classes	Adjacent modality; not equivalent to whole-shrimp farm photographs

Table 3.4 separates reusable data resources from study-specific collections. ShrimpDiseaseImageBD is the most direct public classification source, while TigerShrimpBD supports additional source diversity. The absence of public, expert-reviewed BG/WSSV instance masks explains why the segmentation dataset had to be created by the project. It also means that segmentation results should be interpreted as evidence on a project-specific annotation protocol, not as validation against a recognized clinical mask benchmark.

Table 3.5. Research coverage of representative shrimp vision systems

Study	Core task coverage	Robustness and XAI	Deployment and evaluation validity
Duong-Trung et al. [11]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: No	Mobile/edge: No; leakage control: No; external evaluation: No
Wang et al. [12]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: No	Mobile/edge: No; leakage control: No; external evaluation: No
Querol et al. [13]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: Yes	Mobile/edge: Yes; leakage control: No; external evaluation: No
SHRIMPAI [19]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: No	Mobile/edge: Yes; leakage control: No; external evaluation: No

Study	Core task coverage	Robustness and XAI	Deployment and evaluation validity
Büyükarıkan [23]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: No	Mobile/edge: No; leakage control: No; external evaluation: No
Aung et al. [24]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: No	Mobile/edge: No; leakage control: No; external evaluation: No
Raj et al. [25]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: No	Mobile/edge: No; leakage control: No; external evaluation: No
ShrimpXNet [29]	Classification: Yes; instance segmentation: No	Robustness: Yes; XAI: Yes	Mobile/edge: No; leakage control: No; external evaluation: No
FeatherNetX [17]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: Yes	Mobile/edge: No; leakage control: No; external evaluation: Yes
Explainable ensemble [31]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: Yes	Mobile/edge: No; leakage control: No; external evaluation: No
IoT–ResNet system [27]	Classification: Yes; instance segmentation: No	Robustness: No; XAI: No	Mobile/edge: Yes; leakage control: No; external evaluation: No
Improved YOLO studies [20], [21]	Classification: No; instance segmentation: No	Robustness: No; XAI: No	Mobile/edge: No; leakage control: No; external evaluation: Yes

In Table 3.5, “No” means that the reviewed publication did not report or evaluate the corresponding component; it does not prove that the implementation could not support it. The coverage pattern is nevertheless clear. Classification is common, while mask-level instance segmentation, explicit leakage control, and external evaluation are uncommon. Robustness and XAI appear in a small number of recent studies, and mobile deployment is usually demonstrated in WSSV-specific or system-oriented work rather than in a four-class classification-plus-segmentation framework.

6. Gaps in the Literature/Technology

The first gap is the separation between image-level classification and disease-region localization. Existing shrimp studies predominantly predict a whole-image label or a bounding box. They rarely provide instance masks for BG and WSSV regions, and public datasets do not include independently reviewed mask annotations. As a result, classification confidence can be high without demonstrating that the model has localized visible disease evidence.

The second gap concerns evaluation validity. Most shrimp image papers report image-level splits without a detailed audit of repeated specimens, near duplicates, source overlap, or acquisition clusters. This omission is serious because several photographs can depict the same physical shrimp. A random split may therefore measure recognition of specimen-specific appearance rather than generalization to unseen shrimp. The broader leakage literature shows that subject and site confounding can substantially inflate test performance [89–92].

The third gap is incomplete robustness evidence. Some systems use augmentation or adversarial training, but few evaluate a controlled set of task-appropriate corruptions and report class-balanced performance. The absence of such tests is especially important for mobile farm images, where illumination, focus, compression, background complexity, and camera characteristics can differ from the training set. Corruption results should still be bounded carefully because no finite benchmark can represent every farm and acquisition condition.

The fourth gap is weak comparability across studies. Reported metrics frequently use different disease spaces, datasets, image modalities, split policies, and averaging rules. Accuracy is often presented without Macro-F1 or classwise evidence, which can conceal poor minority-class behavior. Model size, latency, and deployment feasibility are also reported inconsistently. A defensible project must therefore establish controlled internal baselines rather than select a method from headline accuracy alone.

The fifth gap concerns reproducibility and artifact traceability. Several recent systems describe complex preprocessing, augmentation, ensembles, or optimization pipelines without releasing a complete executable package. When code, exact splits, class order, checkpoint-selection rules, and environment versions are unavailable, a reimplementation can legitimately differ from the paper result. Such differences must be analyzed through data protocol, preprocessing, augmentation, architecture, training, model selection, evaluation, and environment factors rather than attributed to one vague cause.

The sixth gap is the limited integration of research and application evidence. Mobile studies demonstrate useful interfaces but usually focus on binary WSSV screening, while multi-class studies often remain offline. Instance segmentation, robustness, XAI, Firebase-backed application workflows, and source-aware external evaluation are generally developed in separate publications. No reviewed system reports all of these components under a single traceable project protocol.

The seventh gap is annotation quality for disease-region segmentation. Producing masks requires both identifying the disease category and deciding where visible evidence begins and ends. This is substantially harder than assigning a whole-image classification label. Background removal may help classification by suppressing irrelevant context, but the same preprocessing can remove spatial or boundary information needed for segmentation. Single-annotator labeling improves internal consistency but leaves uncertainty because inter-annotator agreement and expert mask review are unavailable.

7. Justification for the Project

The literature justifies a project centered on lightweight four-class classification and BG/WSSV instance segmentation rather than object detection. Classification provides an efficient preliminary screening mechanism for Healthy, BG, WSSV, and co-infection images. Instance segmentation supplies a stricter, spatially explicit task that cannot be replaced by classification confidence or a bounding box. The two tasks therefore address different evidence requirements within the same application domain.

The classification direction is justified by the trade-off exposed in existing systems. High-capacity models and ensembles can report strong results but may be unsuitable for mobile inference, while compact systems can sacrifice class-balanced performance. Benchmarking representative TIMM and official YOLO classification families under one controlled seed and split provides an engineering basis for selecting a lightweight backbone. ShrimpXNet remains the closest published methodological reference, but the proposed classifier is evaluated as a direct improvement to the selected YOLO26m-cls engineering baseline rather than as a modification of ShrimpXNet.

The segmentation direction is justified by the scarcity of shrimp disease masks and the limitations of image-level explanation. Published attention and instance segmentation methods provide foundations, but the project-specific attention configuration and Fourier experiments are evaluated as proposed research. The Fourier study directly tests whether frequency-domain transformations improve robustness, small-region evidence, or instance segmentation performance across multiple configurations. Its conclusions must be based on the recorded experiments rather than on theoretical expectation alone.

Leakage-aware evaluation is justified by repeated-specimen risk. Grouping known views of the same shrimp protects the segmentation test set from one important source of information overlap. Source-wise splitting before multi-dataset merging similarly prevents images from being moved across partitions after combination. These measures do not guarantee universal generalization, but they provide a more credible estimate than an undocumented image-level random split.

Finally, mobile integration is justified as an application layer after the research methods are evaluated. TensorFlow Lite conversion, Android CPU inference, result and mask display, history management, Firebase services, and administrator functions demonstrate that selected models can participate in a usable workflow. The application remains a preliminary screening prototype: it does not replace laboratory diagnosis, veterinary assessment, treatment prescription, or formal farmer user-acceptance testing.

Taken together, the project is necessary because the reviewed literature rarely combines lightweight classification, class-imbalance handling, robustness testing, XAI, mask-level disease localization, leakage-aware splitting, multi-source evaluation, reproducibility artifacts, and mobile deployment. The intended contribution is not to claim superiority over incomparable published percentages. It is to construct and evaluate a traceable research pipeline in which each methodological claim is tied to a controlled baseline, a documented dataset protocol, and explicit limitations.